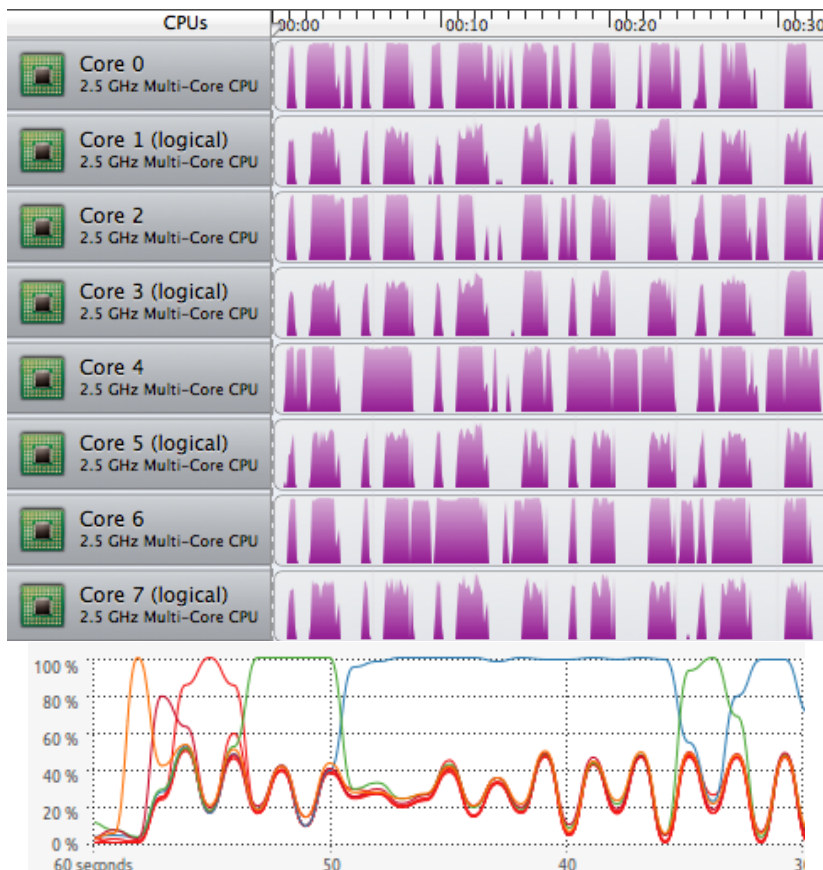
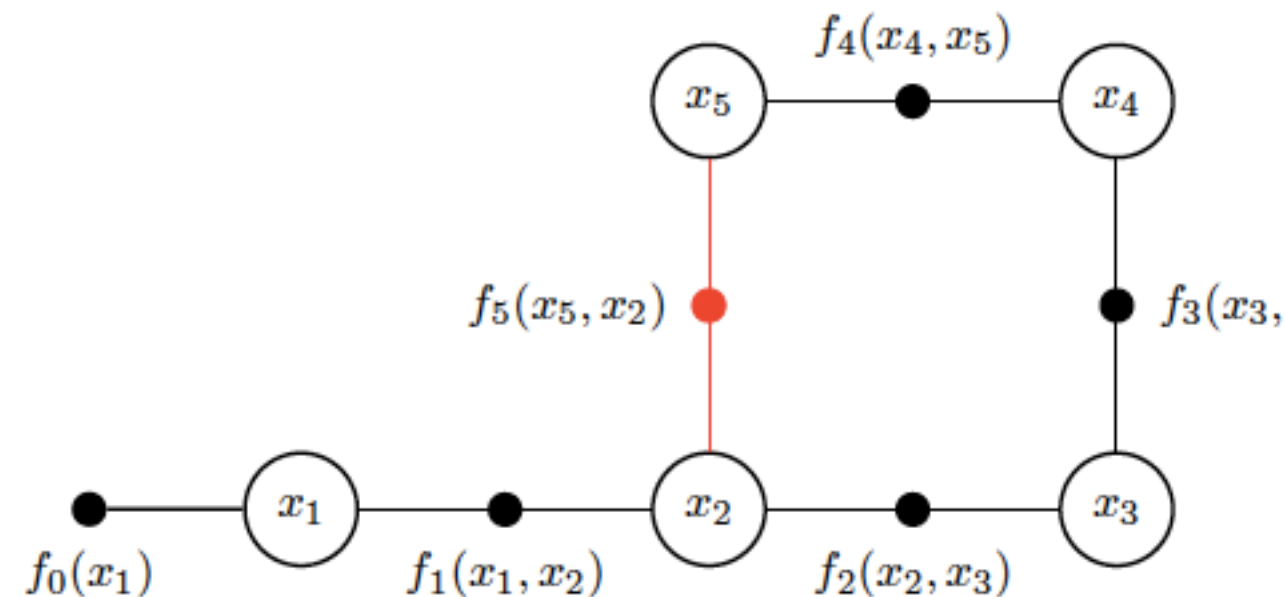


GTSAM: Georgia Tech Smoothing and Mapping

- C++ library
- Open, BSD-licensed: gtsam.org
- Does SFM, SLAM, iSAM, iSAM2...
- Based entirely on Factor Graphs:
- Prototyping in MATLAB:
- Multi-threaded:



```
1 NonlinearFactorGraph graph;
2 noiseModel::Diagonal::shared_ptr priorNoise =
3     noiseModel::Diagonal::Sigmas(Vector_(3, 0.3, 0.3, 0.1));
4 graph.add(PriorFactor<Pose2>(1, Pose2(0, 0, 0), priorNoise));
5
6 // Add odometry factors
7 noiseModel::Diagonal::shared_ptr model =
8     noiseModel::Diagonal::Sigmas(Vector_(3, 0.2, 0.2, 0.1));
9 graph.add(BetweenFactor<Pose2>(1, 2, Pose2(2, 0, 0), model));
10 graph.add(BetweenFactor<Pose2>(2, 3, Pose2(2, 0, M_PI_2), model));
11 graph.add(BetweenFactor<Pose2>(3, 4, Pose2(2, 0, M_PI_2), model));
12 graph.add(BetweenFactor<Pose2>(4, 5, Pose2(2, 0, M_PI_2), model));
13
14 // Add pose constraint
15 graph.add(BetweenFactor<Pose2>(5, 2, Pose2(2, 0, M_PI_2), model));
```

OSPO Summer internship

- Required:
 - python proficiency
 - communication skills
- Gravy:
 - GTSAM experience
 - C++
- Opportunities:
 - Evangelism
 - Documentation
 - Show off applications
 - Vibe-coding vizualizations